# Simulation-informed transfer learning improves low-data nanobody thermal stability prediction

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Experimental melting-temperature (Tm) measurements are scarce for nanobodies, but related stability labels can be generated at larger scale by computation. These labels are not interchangeable with Tm: a mutation free-energy change is a relative, per-mutation quantity, whereas Tm is an absolute experimental phenotype. We tested whether such computed labels can nevertheless improve low-data Tm prediction when they are used as source tasks and models are selected only by experimental Tm validation. On a public benchmark with 57 training, 114 validation, and 396 held-out test sequences, we trained multi-task models that shared an ESM2 sequence representation between a Tm predictor and one computed source task at a time. Source labels included alchemical free-energy perturbation (FEP) mutation free energies, structure-based mutation scores, Rosetta scores for language-model-proposed variants, and molecular-dynamics trajectory summaries. FEP labels gave the largest and most consistent transfer, reducing held-out Tm mean absolute error from 6.61 to 6.26°C, with a paired improvement of 0.35°C (90% confidence interval, 0.24–0.46°C). The effect was source-specific: an MD-derived Q-value did not improve Tm prediction, and larger ESM2 encoders did not reproduce the FEP gain. Rosetta-scored proposal sets gave smaller gains, with no resolved advantage over random variants. These results show that computed stability labels can aid nanobody Tm prediction when their physical meaning transfers to the experimental endpoint.

nanobody thermostability | melting temperature | simulation-to-real transfer learning | protein language model | free energy perturbation

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## Introduction

Machine learning can screen molecular and materials spaces before experimental testing, but its accuracy is limited by the labels used for training. In materials informatics, this label bottleneck has motivated simulation-to-real (Sim2Real) strategies: models are trained on large computational databases from first-principles calculations, quantum chemistry, or molecular dynamics, and are then transferred to smaller experimental target sets (1–3). For polymers and inorganic materials, the transferred error can fall predictably as the computational database grows, which gives a concrete way to judge whether more simulation is worth generating (3). The premise matters because simulation data are not useful by default: model assumptions, finite sampling, and systematic domain gaps separate any calculation from the measurement it stands in for.

Protein engineering has the same bottleneck, but the transfer problem is often less direct. Experimental measurements are expensive and sparse, whereas simulations and physics-based calculations can label many designed or mutated sequences. The calculated quantity, however, is usually not the experimental target. A protein's melting temperature (Tm), the absolute temperature at which half of the molecules are unfolded under a given assay, is one such target. What can be computed or measured at scale is usually adjacent to Tm: a mutation free-energy change ($\Delta\Delta G$), a structure-based stability score, or a summary of a molecular-dynamics trajectory. A $\Delta\Delta G$ value is relative and per-mutation, so it cannot be assumed to improve prediction of an absolute Tm measured on different sequences. The central question is therefore which simulation-derived quantities transfer to the experimental phenotype.

We study this question in nanobodies. These single-domain VHH antibodies are small, modular binding scaffolds (4, 5)—but their use in expression, storage, and formulation hinges on thermal stability (6–8), and measuring or simulating that stability directly is costly (9–12). Sequence-based predictors built on protein language models (PLMs) provide one approach to this problem. Models such as ESM2 learn general-purpose representations from large sequence corpora (13–15), and several methods now predict Tm from such representations (16–21). High-throughput assays are reshaping the data landscape as well: Tsuboyama et al. used cDNA-display proteolysis to measure folding stability for roughly 776,000 natural and designed domains (22). That resource does not yet cover nanobodies. The curated domains span 40–72 residues, whereas VHH domains are longer, and comparable high-throughput $\Delta G$ data are not yet available for them; a model fine-tuned on the mega-scale set also transferred less well to 177–501-residue proteins than to small domains (23). Large stability resources are therefore valuable, but target-specific supervision and validation remain essential for nanobody Tm prediction (24–26).

Here we test whether simulation-derived labels can supply that supervision when experimental Tm is scarce. We treat Tm prediction as the target task and each computational label as a source task that shapes a shared sequence representation, with ESM2 as the encoder (14). The source labels span several kinds of information: mutation free energies from alchemical free-energy perturbation (FEP), Rosetta mutation scores, ThermoMPNN stability scores, Rosetta scores on variants proposed by ESM2 or drawn at random, and an MD-derived Q-value that tracks native-contact persistence. To keep the comparison target-centered, every model setting is chosen on an experimental validation set, and every final number is reported on the same held-out experimental Tm test set using paired error comparisons. This design separates improved source-task learning from improved Tm prediction.

The transfer is source-dependent. FEP mutation free energies, although relative and per-mutation, improve absolute Tm prediction over training on Tm alone, whereas the MD-derived Q-value does not under the same protocol. Rosetta-scored ESM2 proposals give a weaker exploratory signal, suggesting a possible direction for future generator–physics–predictor design loops without demonstrating closed-loop optimization here. Together, these results extend the Sim2Real argument to a protein setting in which source and target labels differ in physical meaning: computed data are useful when the quantity they encode transfers to the experimental property of interest.

## Materials and Methods

### Experimental and technical design

The study tests whether a computational source label can improve experimental nanobody Tm prediction when that label is physically related to Tm but not identical to it. We used a multi-task transfer-learning design: experimental Tm prediction was the target task, and each computational label was a source task that shared part of the network with the target. Throughout, model selection and every reported number were tied to experimental Tm performance. Source-task losses shaped the shared representation during training, but source-task validation errors were not used to pick checkpoints or hyperparameters.

### Experimental Tm data set and split definitions

Experimental Tm labels came from the public nanobody benchmark NbBench, which provides a thermo-seq task with predefined splits (25). We deliberately assigned the smallest partition to training and the largest to the held-out test set, giving 57 training, 114 validation, and 396 test sequences. This split places the model in the intended low-data regime while retaining a large test set for paired error comparisons and label-count analyses.

Split definitions. The training split fit model parameters. The validation split determined checkpoint selection, early stopping, and hyperparameter choice. The test split was used only after settings were fixed. The same rule was applied to Tm-only models and to multi-task transfer models.

Target-label scaling. The target CSV files held an amino-acid sequence column and a Tm label in degrees Celsius. During training, Tm labels were mapped to a scaled regression target: a robust scaling step followed by min–max scaling to [0,1], with the scaler fit only on the 57 training labels. The fitted scaler was then applied to validation labels during training. At evaluation, predictions were inverse-transformed back to degrees Celsius before errors were computed, so the validation and test label distributions never entered the scaling transform.

Experimental-label scaling reference. For this reference, the Tm training split was subsampled to the requested number of labels using the run seed, while the validation and test splits were left intact. These runs give a target-label baseline for judging whether computational source-label scaling moves in the expected direction.

### Computational source-label data sets

Computational labels were loaded as source tasks rather than as additional Tm measurements.

Source-label classes. The source labels were mutation free-energy changes from alchemical free-energy perturbation (FEP), structure-based mutation-effect scores from Rosetta, learned mutation-effect scores from ThermoMPNN, Rosetta scores on ESM2-proposed and random variants, and an MD-derived Q-value summarizing native-contact persistence. All source labels were preprocessed to [0,1]-scaled regression targets and entered the model through source-task heads.

Processed source-table sizes. Mutation-effect sources were organized as two template-specific tables derived from the 1MEL and 4IDL nanobody structures. FEP, Rosetta, and ThermoMPNN each contained 435 and 409 processed rows for the two templates. ESM2-proposed variants and random variants scored by Rosetta each contained 1000 and 1000 processed rows. The MD-derived Q-value source came from a separate panel of simulated single-domain antibody structures drawn from SAbDab, the Structural Antibody Database (27); its processed table contained 1143 analysis-ready rows. Row counts exclude headers.

Sequence-overlap check. We checked exact sequence overlap between the source-label tables and the NbBench splits. The mutation-effect tables and the Rosetta-scored variant tables, which carry explicit sequence columns, had no exact match to any NbBench training, validation, or test sequence. The SAbDab-derived MD table contained a few exact matches, including eight test-set sequences. Because the MD Q-value source did not improve final Tm prediction, this overlap cannot account for the positive FEP result; we report it for transparency.

Source-label sampling. For a mutation-effect experiment with requested count $n$, up to $n$ rows were sampled independently from each of the two template tables using the run seed, and the two tables were given separate task identifiers. For an MD-derived Q-value experiment with count $n$, $n$ rows were sampled from the single Q-value table. Sampled rows were split 80/20 into source-training and source-validation rows with the same seed. Source-validation rows were retained for monitoring and control analyses; in the reported protocol, checkpoint selection and early stopping used only the experimental Tm validation rows.

### SAbDab-derived MD simulations and Q-value labels

The MD source labels were built from nanobody structures selected from a SAbDab-derived summary table (27). We took one heavy-chain nanobody structure per PDB entry when an IMGT-renumbered coordinate file was available and the chosen chain could be verified in that file. The 400 K simulation panel held 1146 nanobody systems. Requiring a usable trajectory, topology, reference structure, a sequence of at least 50 residues, and at least one selected native contact left 1143 trajectories in the processed table.

Each system was prepared from the selected nanobody chain alone. We used the IMGT-renumbered PDB as the structure source, extracted the heavy chain, dropped non-protein components, left the termini uncapped, and assigned protonation at pH 7.4. Structure cleaning and missing-atom handling used PDBFixer, followed by pdb2pqr/PROPKA for pH-aware protonation and Amber-compatible atom naming when available (28, 29). Systems were solvated in explicit OPC water with 15 Å padding and 0.15 M NaCl, and Amber topologies were built with the ff19SB protein force field and OPC water model (30–32).

Trajectories were run with OpenMM on CUDA devices (33). Each system was equilibrated at 300 K with 1 ns NVT and 1 ns NPT at 1 atm, using a 2-fs timestep without hydrogen-mass repartitioning. The 400 K branch was then started from the 300 K equilibrated state through two restrained NVT relaxation stages—1 ns at 400 K with C$\alpha$ restraints, then a second 1 ns 400 K restrained NVT. Production ran 100 ns of restraint-free 400 K NVT with hydrogen-mass repartitioning (4 amu hydrogen mass), a 4-fs timestep, Langevin middle integration at a 1 ps$^{-1}$ collision rate, PME electrostatics with a 1.0-nm nonbonded cutoff, and coordinate/energy output every 100 ps. The MD source labels use this 400 K production branch.

Q-values were extracted from the 400 K production trajectories with MDTraj (34). For each trajectory, the topology was loaded from the Amber parameter file and the native reference was frame 0 of production. A smooth native-contact fraction (35) was computed from backbone-heavy-atom contacts and averaged over the terminal portion of the trajectory, then min–max scaled to [0,1] before training. Supplementary Methods describe the contact selection, contact function, averaging window, and column convention.

### FEP mutation free-energy labels

The FEP source labels were mutation free-energy changes for variants of the 1MEL and 4IDL nanobody templates, computed with the same protocol as our related nanobody thermostability study. Alchemical mutation simulations were run with NAMD 3 using GPU acceleration (36). Input systems were prepared in VMD (37) with a customized AlaScan workflow (38), and coordinates were initialized from AlphaFold 3 models of the wild-type sequences (39). Systems used the CHARMM36 protein force field (40), TIP3P water (41), and 150 mM NaCl.

Each transformation used a dual-topology mutation setup with a CHARMM36 hybrid-residue topology, split into 20 equally spaced $\lambda$ windows from wild type to mutant. Forward and backward transformations were run for 1 ns per direction with a 2-fs timestep and 200 ps of per-window equilibration before data collection, all in the NPT ensemble at 300 K and 1 atm with Langevin temperature control and a Langevin piston barostat. Bonds to hydrogen were constrained with SHAKE (42), and long-range electrostatics used particle mesh Ewald (43). Relative free energies were estimated with the Bennett acceptance ratio (44). The processed tables retain the raw mutation free-energy value, its negated value, and the min–max scaled label used by the network; under this convention, larger processed labels correspond to more favorable mutation free energies.

### Rosetta and ThermoMPNN mutation-effect labels

Rosetta mutation-effect labels were generated with the Rosetta ddg_monomer application (45). For each template, the PDB structure was cleaned to the target chain, variant sequences were compared with the wild-type sequence, and the differences were written to Rosetta mutation-list files. The calculation used full-atom Rosetta with the ref2015 score function (46), 50 iterations, Cartesian minimization enabled, local-only optimization disabled, side-chain-minimization-only disabled, minimum-score aggregation, and no PDB dumping. The same protocol scored the template mutation-effect tables, the ESM2-proposed variants, and the random-variant control.

For the ESM2-proposed source, ESM2 650M sampled 100,000 two-mutation variants from each wild-type template, ranked them by pseudo-log-likelihood, and kept the top 1% for Rosetta scoring (14). The random-variant control used roughly 1000 two-mutation variants with positions and substitutions chosen at random, scored by the same Rosetta workflow.

ThermoMPNN source labels were computed from the same template-specific variant sets with ThermoMPNN, a stability-change predictor trained by transfer learning from larger protein-stability data (47), and converted to the same processed CSV format as the other mutation-effect sources.

### Neural-network architecture

All models used an ESM2 protein language-model encoder (14) with lightweight regression heads. Unless stated otherwise, the encoder was the 8M-parameter ESM2 model. Sequences were tokenized with truncation and padding to 160 residues. The first-token encoder representation fed a shared multilayer perceptron with hidden dimensions 256, 128, and 32, ReLU activations, and dropout. A scalar Tm head predicted the scaled target label; in source-transfer models, additional scalar heads predicted the source labels from the same shared representation.

The Tm-only baseline used the encoder, the shared trunk, and the Tm head with experimental Tm labels alone. Source-transfer models kept this target path and added source heads. The main FEP model used separate source heads for the two template-specific mutation-effect tasks, as selected in the source-head comparison. Controls included a shared mutation-effect head, a shared head conditioned on a template embedding, and a shared head with template-specific affine calibration, along with residual and latent source-fusion variants. The main result used the original shared-trunk architecture.

### Hot-encoder and frozen-encoder training

We evaluated two encoder regimes. In the hot-encoder regime, the ESM2 encoder and the regression heads were optimized jointly, with the encoder given a smaller learning rate than the freshly initialized trunk and heads so that the pretrained representation was updated more gently. In the final FEP setting, the selected encoder learning rate was $3\times10^{-5}$ and the head learning rate was the default $3\times10^{-4}$.

In the frozen-encoder control, all ESM2 parameters were fixed. The encoder was run once to precompute sequence embeddings for the target and source examples, and training then optimized only the trunk and task heads. This control asks whether the source labels still help when the pretrained representation is used as a fixed feature extractor. Encoder-size controls repeated selected Tm-only and FEP settings with 35M- and 650M-parameter ESM2 encoders.

### Loss function and multi-task objective

Each training example was a triple $(x_i, y_i, t_i)$: sequence $x_i$, scaled label $y_i$, and task identifier $t_i$. The target Tm task had one identifier, and each source task had its own. A minibatch could mix target and source examples, but every example contributed loss only through the head for its task.

The per-task regression loss was Huber loss with $\delta = 1.0$ on scaled labels. The default multi-task objective used learnable task-uncertainty weights (48): for task $k$ with loss $L_k$ and learned log-scale $s_k$, the contribution was

$$\frac{L_k}{2\exp(2s_k)} + s_k,$$

and the total loss summed over the tasks present in the minibatch. Fixed source-task weights were also evaluated as controls, particularly for the MD-derived labels.

### Training, early stopping, and hyperparameter selection

Models were trained with AdamW, cosine learning-rate scheduling, batch size 16, weight decay 0.04, 100 warmup steps, and at most 400 epochs. The default head learning rate was $3\times10^{-4}$, the default hot-encoder learning rate was $1\times10^{-4}$, and the default dropout was 0.15. Training used mixed precision on CUDA devices, with checkpoints saved and evaluated once per epoch.

Within each run, the best checkpoint was the one with the lowest mean squared error on the experimental Tm validation examples, and early stopping used the same metric with a patience of 15 epochs. Source-label rows were present in the training set, but were excluded from the validation data used for checkpoint selection and early stopping. Thus, reported checkpoints were selected by target-task validation performance rather than by source-task performance.

Hyperparameters were likewise selected on the experimental Tm validation split. Candidate settings varied the encoder learning rate, head learning rate, dropout, loss-weighting mode, source-head design, and selected architecture controls. The search covered encoder learning rates of $3\times10^{-5}$, $1\times10^{-4}$, and $3\times10^{-4}$; a lower head learning rate of $1\times10^{-4}$ paired with encoder learning rate $3\times10^{-5}$; dropout rates of 0.05, 0.15, and 0.30; and fixed source-task loss weights where relevant. Each candidate was evaluated as a three-seed ensemble on the Tm validation split. The selected setting for each source class was then evaluated on the held-out test split as a ten-seed ensemble.

Independent run seeds controlled model initialization, target-training subsampling in the target-label scaling experiment, source-label sampling, and source-task train/validation splitting. Final comparisons used seeds 1–10. The label-count curves used fixed source-specific training recipes while varying the number of source labels; a separate model was not retuned at each label-count point.

### Final evaluation and statistics

Final evaluation used the same 396 held-out Tm test examples for every condition. For each condition, ten independently seeded models produced scaled Tm predictions for every test sequence; these were averaged in scaled-label space and inverse-transformed to degrees Celsius. The primary metric was mean absolute error (MAE) in degrees Celsius. Root mean squared error, coefficient of determination, and Spearman and Pearson correlations were also computed, but MAE anchored the main comparisons because it reads directly as an average temperature error.

Confidence intervals came from nonparametric bootstrap resampling over held-out test proteins. For a single condition, the absolute-error vector from the ten-seed ensemble was resampled with replacement 10,000 times using a fixed random seed, and the reported 90% interval is the 5th–95th percentile range of the bootstrapped MAE values.

For paired comparisons, the same bootstrap indices were applied to the absolute-error vectors of the candidate and reference conditions, and the candidate-minus-reference MAE difference was computed for each bootstrap sample. This paired bootstrap preserves the matched-test-example structure of the comparison. Throughout, $\Delta$MAE is candidate minus reference, so negative values mean lower error than the reference. The evaluation code also reports the one-sided tail probability—the fraction of paired bootstrap samples with $\Delta$MAE above zero—but we emphasize effect sizes and 90% confidence intervals.

### Implementation details

The pipeline builds the target and source tasks, applies target-label scaling, performs seeded source-label sampling, filters validation data for target-task checkpoint selection, evaluates seed ensembles, and writes bootstrap statistics. Each run emits a machine-readable summary with the command-line arguments, resolved model settings, per-point metrics, bootstrap intervals, and paired comparisons. These summaries feed the figure-generation workflow directly and will be archived with the processed data for reproducibility.

## Results

### A target-centered framework for using simulation labels in Tm prediction

We first defined a target-centered evaluation framework. The target task was nanobody Tm regression from amino-acid sequence, and computational labels were treated as source prediction tasks rather than as substitutes for experimental Tm. Each model used the pretrained protein sequence model ESM2 to encode the sequence, followed by a shared representation used by a Tm head and a computed-label head. Thus, the source task could influence the representation while the final objective remained experimental Tm prediction (Fig. 1).

Every candidate setting was selected on an experimental Tm validation set, and the final numbers came only from a held-out experimental Tm test set; improvements were measured as paired differences in absolute error on the same test examples. This protocol distinguishes a model that learns the computational label from one that improves the experimental endpoint.

### Scaling depends on the physical label

We next varied the number of experimental and computational labels. Adding experimental Tm labels improved the Tm-only model, as expected (Fig. 2). The target-label sweep used 10, 20, 30, 40, and 57 experimental Tm training labels; the FEP sweep used 10, 40, 80, 160, and 320 rows per template-specific source table; and the MD Q-value sweep used 10, 40, 80, 160, 320, and 640 rows from the single MD source table. FEP mutation free-energy labels showed a beneficial trend, with the largest FEP setting giving the lowest held-out Tm error among the FEP points. Because the curve was not strictly monotonic, we do not interpret it as a power law. The supported conclusion is that additional mutation free-energy labels can improve Tm prediction when model selection and evaluation are both based on experimental Tm.

The MD-derived Q-value behaved differently. This label measures how well native contacts persist during high-temperature MD trajectories. Adding more of these labels did not reliably beat learning from Tm measurements alone, even though they also came from simulation. The sweep therefore indicates that label count alone is insufficient; the physical content of the source label matters.

### All-atom simulation labels differ in their transfer to Tm prediction

We then focused on two all-atom source labels. Training on experimental Tm alone reached a held-out test MAE of 6.61°C. Adding FEP mutation free-energy labels reduced this error to 6.26°C, with a paired $\Delta$MAE of -0.35°C (90% confidence interval, -0.46 to -0.24°C; Fig. 3); negative paired changes mean lower error than the Tm-only reference. The MD-derived Q-value increased the test error to 6.73°C (paired change, +0.12°C; 90% confidence interval, +0.01 to +0.22°C). Both labels came from atomistic simulation, but only the mutation free-energy label improved the experimental endpoint.

The encoder controls reinforced this split. With a frozen encoder, errors rose overall, but FEP still beat the frozen Tm-only baseline: 7.20°C versus 7.38°C MAE. The gain was smaller than with a hot encoder, and the frozen FEP model stayed worse than the hot-encoder FEP model. The MD Q-value helped in neither regime. FEP labels therefore carry useful information even when the representation is fixed, while encoder adaptation produced the largest target-task gain.

### The FEP gain is not explained by sequence-model size alone

Because protein language-model size can affect downstream performance, we checked whether a larger encoder alone could reproduce the FEP benefit. It could not. In the Tm-only setting, the 35M and 650M encoders did not beat the 8M encoder on held-out performance. In the FEP setting, the 35M and 650M encoders reached 6.50 and 6.65°C MAE, both worse than the 8M FEP model. FEP labels remained beneficial within each size comparison, but the lowest absolute error came from the smallest encoder paired with FEP labels. Within the tested range, the gain tracked the source label rather

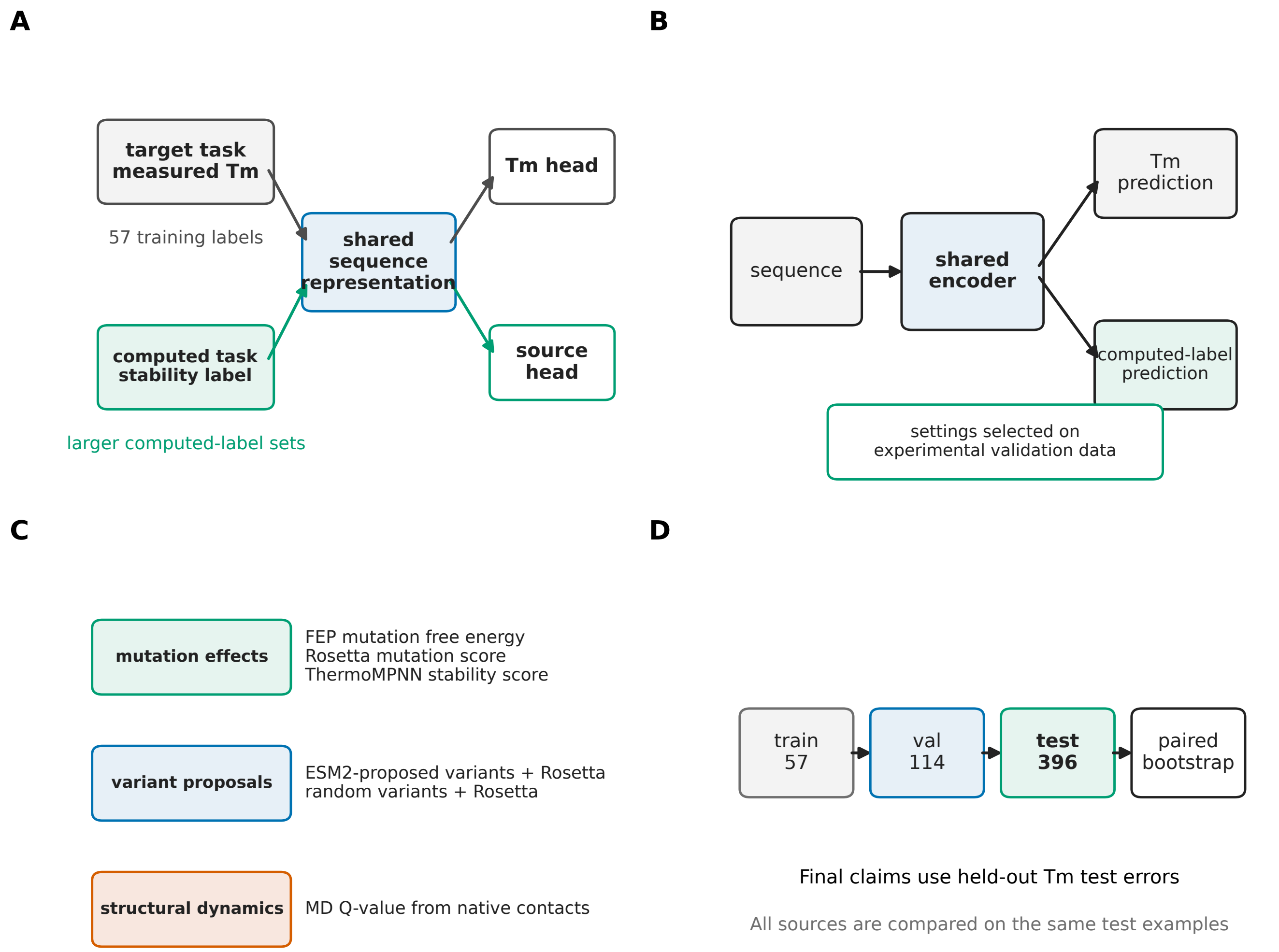


Fig. 1. Multi-task transfer-learning framework for using simulation data in experimental Tm prediction. (a) Experimental Tm labels are scarce, whereas simulation and physics-based calculations can provide larger computed-label pools. (b) Nanobody Tm prediction is used as a concrete case in which the computed labels are related to, but not identical with, the experimental target. (c) The model shares a pretrained protein sequence representation between the Tm target head and a computed-label head. (d) Model settings are selected on an experimental validation set, and final claims are evaluated on held-out experimental Tm test examples using paired error statistics.

than encoder scale (Supplementary Fig. 4).

## Structure-based labels provide weaker source signals

We next looked at computational labels outside the FEP and MD pipelines. Rosetta scores on ESM2-proposed variants reached 6.45°C MAE (paired change, -0.17°C; 90% confidence interval, -0.31 to -0.03°C), and ThermoMPNN stability scores reached 6.46°C MAE (paired change, -0.15°C; 90% confidence interval, -0.30 to -0.01°C; Fig. 4). Rosetta mutation scores reached 6.53°C MAE (paired change, -0.09°C; 90% confidence interval, -0.20 to +0.02°C), and random variants scored by Rosetta reached 6.51°C MAE (paired change, -0.10°C; 90% confidence interval, -0.22 to +0.03°C). All were weaker than FEP, and the two Rosetta controls had confidence intervals that crossed zero.

The ESM2-proposed variants scored by Rosetta slightly outperformed the random variants, with a paired MAE difference of -0.07°C in favor of the ESM2-proposed set. The confidence interval crossed zero, so we read this as exploratory rather than as evidence that the present generator improves variant selection. The result shows that language-model proposals can be annotated by physics-based scoring and used as source labels for the Tm predictor, which we return to in the Discussion.

Across the labels we tested, the strongest and most consistent improvement came from FEP mutation free energies. Other structure-based labels and proposal-set scores gave weaker gains, and the MD-derived Q-value gave none. The pattern is selective rather than additive: a computational label helps only when its information transfers to the experimental Tm endpoint.

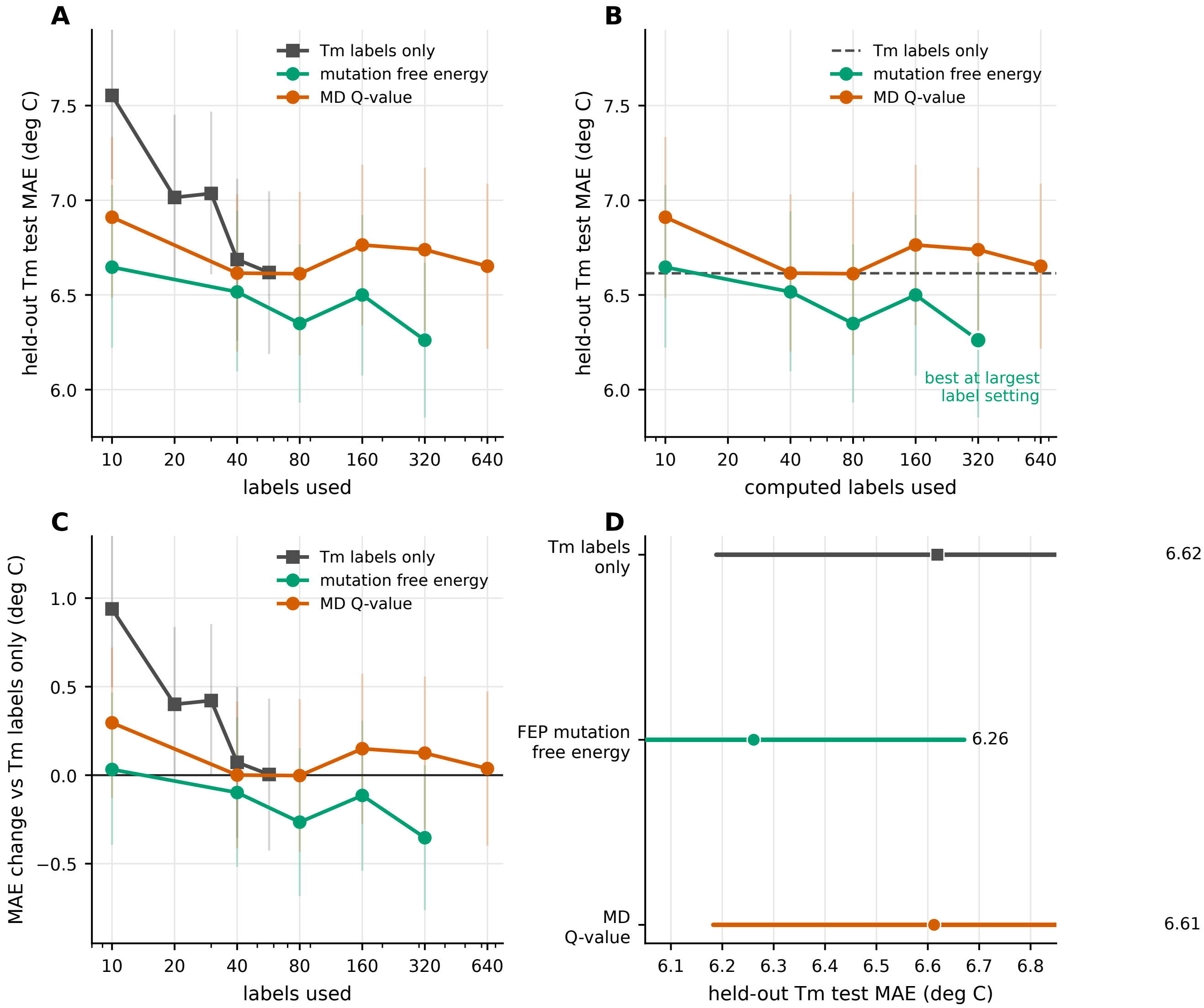


Fig. 2. Label-count scaling of experimental and computational supervision. (a) Experimental Tm-label scaling, FEP mutation free-energy scaling, and MD Q-value scaling are plotted together to compare the direction of label-count effects. (b) Computed-label scaling for FEP and the MD Q-value, with the Tm-label-only reference shown as a dashed line. (c) The same curves shown as MAE changes relative to the Tm-label-only reference. In panels a–c, points show ensemble MAE values and thin vertical intervals show 90% bootstrap intervals. (d) Best points from the label-count sweeps are replotted as interval estimates to compare the experimental-only, FEP, and MD Q-value references.

# Discussion

## Source-label transfer is selective

This study asked how simulation-derived labels should be used when the experimental target is scarce and the computed quantity is related to, but not identical to, that target. In nanobody Tm prediction, FEP mutation free-energy labels produced a modest but consistent improvement in held-out Tm prediction. An MD-derived Q-value did not improve the same endpoint, and increasing the ESM2 encoder size did not reproduce the FEP gain. The result supports a selective view of simulation-to-experiment transfer: computed labels are useful when their physical content transfers to the experimental phenotype under target-centered validation.

## Why mutation free energies transfer

The useful source label did not measure Tm directly. FEP estimates a mutation free-energy change, a relative and per-mutation $\Delta\Delta G$, whereas Tm is an absolute temperature at which folded and unfolded populations balance under a specified assay. These quantities are usually treated as separate prediction targets. They are nevertheless connected through the same stability landscape: Tm measurements anchor absolute stability for observed sequences, while mutation free energies describe local directions in sequence space. The present result shows that this shared physical content is strong enough to cross the task boundary. $\Delta\Delta G$ supervision, with checkpoints selected only by experimental Tm validation, improved Tm prediction on held-out sequences that were not part of the FEP template-variant sets.

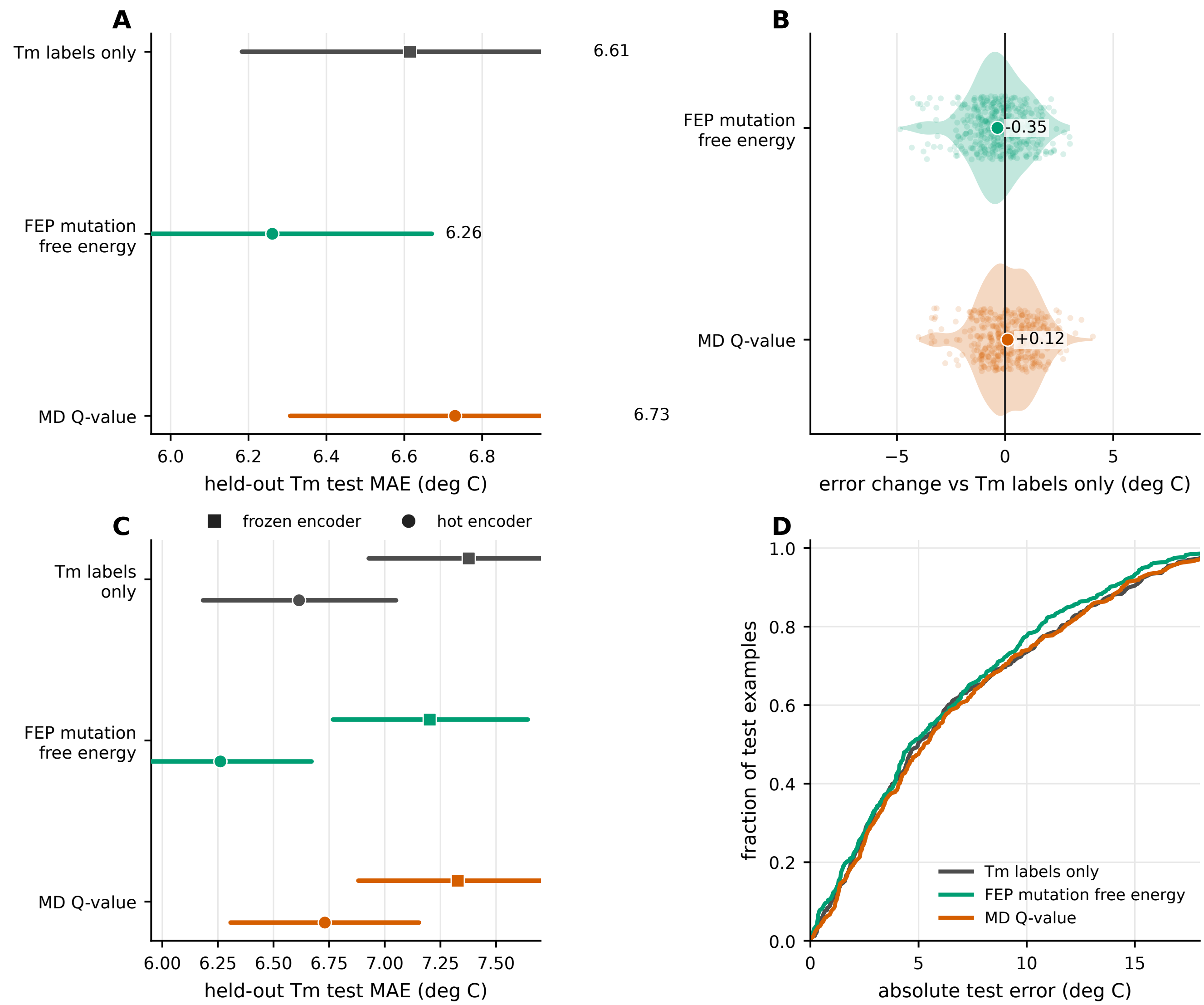


Fig. 3. All-atom simulation labels differ in their transfer to Tm prediction. (a) Final test MAE for experimental Tm labels alone, FEP mutation free-energy labels, and the MD-derived Q-value. Intervals show bootstrap confidence intervals over test-set absolute errors. (b) Per-example and paired changes in absolute error relative to training on Tm labels alone. Negative values indicate lower error than the reference. (c) Frozen-encoder and hot-encoder comparisons for the same three conditions. (d) Cumulative distributions of held-out absolute errors.

This finding is not explained by the presence of an auxiliary task alone. The MD-derived Q-value was also simulation-derived, but it did not transfer under the same protocol. The distinction matters because relative stability labels can be generated in bulk by physics-based calculation, and high-throughput folding stability measurements are now available for large collections of protein domains (22). If such labels are physically matched to the phenotype of interest, they can provide supervision for targets, such as nanobody Tm, for which direct measurements remain limited.

## Controls and interpretation

The controls narrow the interpretation. The FEP label-count sweep was consistent with benefit at the largest setting, but the curve was not monotonic; therefore, this study does not establish a biomolecular scaling law. The conclusion is instead that source-label identity matters. This both parallels and extends the materials-informatics Sim2Real setting, where transferred error can scale with the size of a related computational database (3). Here, the useful source labels were not additional Tm measurements, but mutation free energies.

Encoder controls led to the same conclusion. The 35M and 650M ESM2 encoders did not outperform the 8M encoder in the Tm-only or FEP settings, and the lowest held-out error came from the 8M encoder trained with FEP labels. Frozen-encoder experiments showed that FEP labels still helped when ESM2 was used as a fixed feature extractor, but the largest gain required updating the encoder. These results favor controlled encoder adaptation guided by physically relevant source labels, rather than increasing language-model capacity alone.

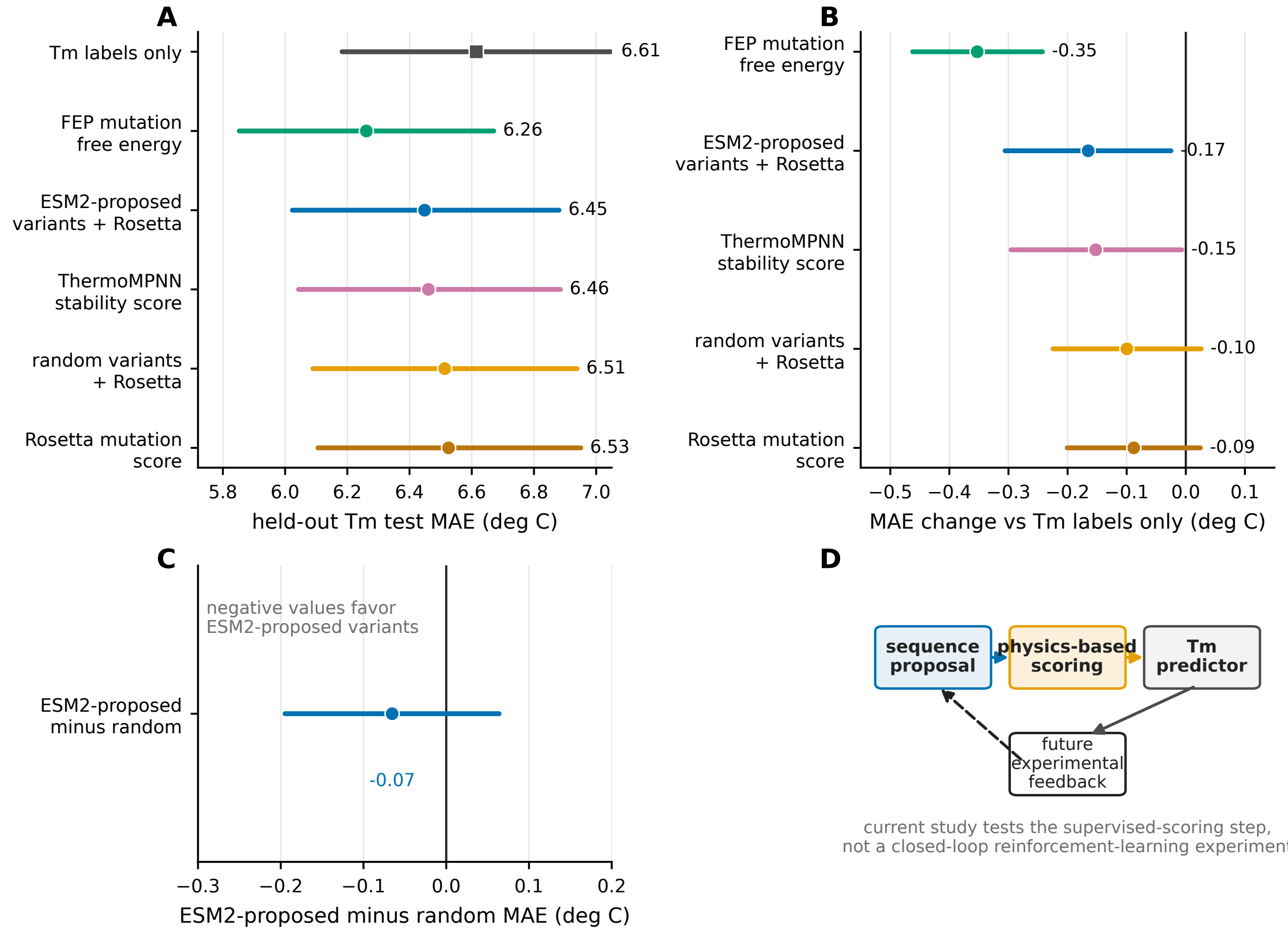


Fig. 4. Structure-based labels and language-model-proposed variants. (a) Final test MAE for Tm-only learning, FEP, and structure-based source labels. (b) Paired ΔMAE estimates relative to Tm-only learning. Negative values indicate improvement. (c) Direct paired comparison between ESM2-proposed variants scored by Rosetta and random variants scored by Rosetta. Negative values favor the ESM2-proposed set. (d) The proposal-set analysis motivates future tests of a generator–physics–predictor loop, but the present study evaluates only the supervised-scoring step and does not perform closed-loop optimization.

The MD result should be read with similar specificity. It does not imply that all MD-derived descriptors are unsuitable for Tm prediction. Rather, the particular high-temperature native-contact summary used here did not transfer. Compressing each trajectory to a single contact-persistence statistic may omit the mutation-specific energetics that determine how sequence changes shift stability. Future MD-based source labels should be designed around observables that are closer to mutation effects or thermodynamic stability, rather than around global structural persistence alone.

### Physics-scored sequence proposals

The Rosetta-scored proposal-set experiment was an exploratory test of a possible design workflow. Rosetta scores on ESM2-proposed variants improved Tm prediction over Tm-only learning, but their direct advantage over random variants was not resolved. We therefore do not conclude that the present generator optimized nanobody thermostability. The result instead establishes a narrower point: language-model proposals can be annotated by physics-based scoring and used as source labels for a Tm predictor.

This supervised-scoring step is only one component of a closed design loop. A full loop would require a generator to propose variants, physics-based calculations to annotate them, a Tm predictor to update estimates of experimental stability, and those estimates to guide subsequent proposals. Such active- or reinforcement-learning workflows, together with experimental testing of newly proposed high-Tm nanobodies, are beyond the scope of this study.

### Implications for data collection

The practical implication is that simulation data should be chosen by physical relatedness to the target phenotype and then validated by held-out experimental performance. For nanobody Tm prediction, the most direct next source pool would be a larger and more diverse set of VHH-length $\Delta\Delta G$ labels. The current FEP labels covered two templates, sufficient to show transfer but not to establish how performance scales with source-pool size. Broader mutation and template coverage would test whether the qualitative benefit observed here becomes a predictable scaling relationship.

Additional experimental Tm measurements remain important because $\Delta\Delta G$ is relative. A few hundred Tm labels distributed across the relevant stability range would help set the absolute scale that mutation free energies alone cannot define. This combination—broad relative stability labels and a smaller set of absolute Tm anchors—is a natural data design for low-data Tm prediction.

The choice of source label also has a cost dimension. Learned stability predictors and structure-based scores can label large variant pools at much lower cost than all-atom alchemical FEP, and they produced smaller but detectable signals in this study. FEP is more expensive per mutation, but gave the largest per-label transfer. A tiered strategy is therefore plausible: use inexpensive learned or structure-based labels for breadth, and reserve all-atom free-energy calculations for sequence regions or accuracy regimes where higher-fidelity labels are needed. Near the operating point of this study, the experimental-label and FEP label-count curves suggest that one experimental Tm label corresponds roughly to tens of FEP labels, whereas the MD-derived Q-value had an equivalent sample size near zero. Because the FEP curve was not monotonic, this estimate should be treated as a coarse guide rather than a fixed conversion factor.

### Limitations

Several limitations bound the interpretation. The experimental Tm training set is small, and the conclusions should be retested on other nanobody collections and protein families. The source-label comparison covers only the labels available here; other free-energy protocols, stability predictors, or trajectory descriptors may transfer differently. The FEP source pool was too small to establish a scaling exponent, and no new experimental measurements were made to validate high-stability candidates. Finally, raw trajectories are large, so full reproducibility will depend on processed data, analysis code, and a complete simulation protocol.

### Outlook

For low-data protein-property prediction, simulation labels should be treated as targeted sources of physical supervision rather than generic data augmentation. The workflow is to define the experimental phenotype, choose computed labels whose physical meaning is expected to transfer, and judge the combined model only on held-out experimental data. In this study, mutation free energies met that criterion for nanobody Tm. This provides a practical basis for moving from simulation-assisted prediction toward future closed-loop protein engineering.

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AUTHOR CONTRIBUTIONS
Taihei Murakami and Kentaro Sasaki contributed equally to this work. Yasuhiro Matsunaga conceived and supervised the study, acquired funding, developed the analysis and training code, performed the analyses, interpreted the results, prepared the figures, and wrote the manuscript. Kentaro Sasaki performed computational calculations. Taihei Murakami contributed to draft writing. All authors reviewed and approved the final manuscript.

## Declaration of competing interest

Taihei Murakami is an employee of Epsilon Molecular Engineering, Inc. Yasuhiro Matsunaga is a scientific advisor of Epsilon Molecular Engineering, Inc.

## Data availability

Processed data required to reproduce the main and supplementary figures will be deposited in Zenodo at ZENODO_DATA_DOI_PLACEHOLDER. Raw simulation trajectories are large and will be made available at RAW_DATA_LOCATION_OR_REQUEST_POLICY_PLACEHOLDER.

## Code availability

Analysis, training, and figure-generation code will be available on GitHub at GITHUB_REPOSITORY_URL_PLACEHOLDER. An archival release will be deposited in Zenodo at ZENODO_CODE_DOI_PLACEHOLDER.

## AI-use disclosure

AI-assisted tools were used to support language editing, code development, and figure preparation. The authors reviewed and take responsibility for all manuscript content, analyses, and conclusions.

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## Supplementary Methods

### Experimental target data and target-label scaling

All target-task comparisons use the same NbBench thermo-seq endpoint (25). The processed tables hold an amino-acid sequence field and a melting-temperature label in degrees Celsius. The split sizes are 57 training, 114 validation, and 396 held-out test sequences. As described in Methods, the smallest NbBench partition was assigned to training and the largest to testing to create a low-data target setting with a fixed held-out evaluation set.

Split definitions. The splits are fixed throughout. The 57 training examples fit network parameters, the validation split drives checkpoint selection, early stopping, and choice among candidate settings, and the test split stays untouched until those choices are fixed. The same rule applies to Tm-only models and to models trained with computational source labels.

Target-label scaling. Before optimization, the training workflow maps the experimental Tm values to a scaled regression target. The scaler is fit only on the 57 training labels, using a robust scaling step followed by min–max scaling to [0,1]. Validation labels are transformed with this training-fitted scaler during model selection. Test predictions are averaged across seeds in scaled-label space and inverse-transformed to degrees Celsius before errors are computed, so the validation and test label distributions never influence the fitted transform.

Experimental-label scaling reference. This reference uses the same training split but subsamples it to the requested number of labels with the run seed; the validation and test splits are not subsampled. It serves as a positive control for the pipeline, because additional target labels should lower Tm prediction error.

### Processed computational source-label tables

Computational labels are loaded as source tasks rather than appended to the Tm data as additional measurements. Table 1 lists the processed target and source tables used in the reported comparisons. Mutation-effect tables carry sequence fields and [0,1]-scaled source-label fields. The MD-derived Q-value table uses the same processed format so that it can enter the same source-task loader; there, the scaled label is a min–max scaled Q-value rather than a mutation free energy.

Table 1. Processed target and source-label data. Row counts exclude header rows.

| Data type | Role | Processed rows | Use in reported comparisons |
|---|---|---|---|
| Experimental Tm | Target train | 57 | Model fitting |
| Experimental Tm | Target validation | 114 | Checkpoint and setting selection |
| Experimental Tm | Target test | 396 | Final held-out evaluation |
| FEP mutation free energy | Source tasks | 435 + 409 | Main source task |
| Rosetta mutation score | Source tasks | 435 + 409 | Source comparison |
| ThermoMPNN stability score | Source tasks | 435 + 409 | Source comparison |
| Random variants scored by Rosetta | Source tasks | 1000 + 1000 | Source comparison |
| ESM2-proposed variants scored by Rosetta | Source tasks | 1000 + 1000 | Source comparison |
| MD-derived Q-value | Source task | 1143 | Source comparison and scaling control |

Source-task assignment. The loader assigns each row to a target or source task. Experimental Tm rows use the Tm head. In the main mutation-effect model, the two template-specific tables use separate source heads, and the MD-derived Q-value table uses a single source head. This assignment fixes which regression head an example uses and which task-specific loss term it contributes to.

Sequence-overlap check. We checked exact sequence overlap between the processed source tables and the fixed NbBench splits. The FEP/Rosetta mutation-effect tables, the random variants scored by Rosetta, and the ESM2-proposed variants scored by Rosetta had no exact match to any NbBench training, validation, or test sequence. The ThermoMPNN tables used the same template-specific variant design as the mutation-effect sources. The SAbDab-derived MD Q-value table contained exact matches to 2 NbBench training, 4 validation, and 8 test sequences. We disclose this because it bears on interpretation of the MD control; it cannot explain the positive FEP result, and the MD Q-value source did not improve held-out Tm prediction.

Source-label sampling. For mutation-effect experiments with requested count $n$, the loader samples up to $n$ rows independently from each of the two template tables using the run seed; the final FEP setting with $n = 320$ therefore supplies 320 sampled 1MEL rows and 320 sampled 4IDL rows before the source-task split. For MD-derived Q-value experiments, the loader samples $n$ rows from the single Q-value table. Source rows are split 80/20 into source-training and source-validation rows with the same seed. In the reported training protocol, checkpoint selection and early stopping depend only on the experimental Tm validation examples.

### SAbDab-derived nanobody structure panel for MD

The MD source-label panel was built from SAbDab, the Structural Antibody Database (27). The archived SAbDab nanobody summary table holds 2422 rows with PDB identifier, heavy- and light-chain assignments, model index, antigen annotation, experimental method, resolution, species, and other fields. The manifest workflow matches this summary against the archived IMGT-renumbered PDB files.

Structure-selection rule. The selection rule is deterministic. For each PDB identifier with both a SAbDab summary row and an IMGT-renumbered PDB file, the workflow takes the first heavy-chain row in the SAbDab table and accepts the chain only if that chain identifier is present in the IMGT PDB file. The output manifest holds one entry per PDB identifier, with fields for the manifest row number, PDB identifier, selected heavy chain, SAbDab TSV row, IMGT PDB path, and MD run directory. This initial one-per-PDB manifest contained 1177 structures.

400 K simulation panel. The 400 K campaign reused the systems prepared in the 300 K campaign and built a 1146-entry 400 K manifest, consisting of 798 X-ray, 345 electron-microscopy, and 3 NMR structures according to the SAbDab method field. None of the selected rows had an annotated light chain, consistent with a single-domain antibody panel. After trajectory and Q-value processing, 1143 unique PDB identifiers yielded usable source-label rows.

### MD system preparation and simulation protocol

System preparation. Each structure was prepared as a nanobody-only system. The workflow takes the IMGT-renumbered PDB as the structure source, extracts the selected heavy chain, keeps only protein atoms, and leaves the termini uncapped, at pH 7.4. Chains are cleaned with PDBFixer, and pH-aware protonation is assigned through pdb2pqr/PROPKA when available (28, 29), with a fallback to AmberTools-based hydrogen assignment and Amber naming. The prepared chain is solvated in explicit OPC water with 15 Å padding and 0.15 M NaCl, and Amber topology and coordinate files are built with ff19SB and OPC (30–32).

300 K equilibration branch. The base 300 K branch uses OpenMM on CUDA devices (33): 1 ns of NVT followed by 1 ns of NPT at 300 K and 1 atm, with a 2-fs timestep and no hydrogen-mass repartitioning. Production at 300 K was generated for the broader campaign, but the source label used here comes from the 400 K branch.

400 K production branch. The first 400 K relaxation stage restarts from the 300 K equilibrated state and runs 1 ns of NVT at 400 K with C$\alpha$ restraints; the second repeats 1 ns of restrained 400 K NVT. Both relaxation stages use a 2-fs timestep and no hydrogen-mass repartitioning. Production runs 100 ns of restraint-free 400 K NVT, with hydrogen-mass repartitioning, 4 amu hydrogen mass, a 4-fs timestep, and coordinate/energy output every 100 ps.

Integrator and reporting settings. In OpenMM, explicit-solvent periodic systems use PME electrostatics with a 1.0-nm nonbonded cutoff and constraints on bonds to hydrogen. NPT stages use a Monte Carlo barostat; NVT stages omit it. The integrator is the Langevin middle integrator with 1 ps$^{-1}$ friction. The 100-ps reporting interval gives 1000 expected frames for a complete 100 ns production trajectory.

### MD-derived Q-value extraction

Trajectory input requirements. The processed MD table was generated from the archived 400 K production trajectories. A system was eligible only if it had a production trajectory, a prepared reference structure, and an Amber parameter file. The amino-acid sequence was read from the prepared structure using the first C$\alpha$ atom of each residue, and sequences shorter than 50 residues were excluded.

Native-contact definition. Q-values are computed with MDTraj (34). The topology is loaded from the Amber parameter file, and the native reference is frame 0 of production. The calculation uses protein backbone heavy atoms. A candidate native contact must span more than three residues, have a reference-frame distance below 0.45 nm, and include at least one atom from a hydrophilic residue (Asp, Glu, Gln, Asn, Arg, Lys, or His, including Amber protonation variants). For a native-contact pair $(i,j)$ with reference distance $r_{ij}^0$ and instantaneous distance $r_{ij}$, the per-frame contribution is

$$q_{ij} = \frac{1}{1+\exp\{\beta(r_{ij} - \lambda r_{ij}^0)\}},$$

with $\beta = 50$ nm$^{-1}$ and $\lambda = 1.8$, following the smooth native-contact form of Best–Hummer-style Q-value analyses (35). The frame-level Q-value is the mean of $q_{ij}$ over all selected native contacts.

Trajectory-level Q-value label. The trajectory-level label is the mean Q-value over the terminal portion of production. For trajectories with more than 300 frames, the implementation uses the larger of 300 frames and one third of the available frames; for shorter trajectories it uses all frames. For a complete 100 ns trajectory with 100-ps output, this is 333 terminal frames, roughly the final 33 ns. In the processed 400 K table, the number of frames used ranges from 66 to 333 (median 333), the number of native contacts from 28 to 728 (median 173), and the sequence length from 58 to 461 residues (median 122). Raw Q-values range from 0.0437 to 0.9994 and are min–max scaled to [0,1] for source-task training.

### Sequence tokenization and encoder inputs

Sequences are tokenized with the ESM2 tokenizer for the selected encoder (14), using truncation and fixed-length padding to 160 residues. In the production protocol, the trainer receives all target and sampled source rows for training, but its evaluation data are filtered to the experimental Tm validation rows.

Hot-encoder input path. The hot-encoder setting keeps tokenized sequences in the training data and updates the ESM2 weights. The frozen-encoder setting runs ESM2 once per target and source sequence to precompute first-token embeddings, after which the optimizer updates only the regression trunk and task heads.

### Neural-network architecture

The default encoder is the 8M-parameter ESM2 model, and the model uses its first-token representation as the sequence-level embedding. The shared trunk has dimensions hidden-size $\rightarrow 256 \rightarrow 128 \rightarrow 32$, with ReLU activations and dropout after the first two hidden layers. The Tm head maps the 32-dimensional representation to one scaled scalar prediction.

Target and source heads. The Tm-only baseline is the encoder, the shared trunk, and the Tm head. Source-transfer models keep this target path and add source heads. The main mutation-effect model uses separate source heads for the two template-specific tasks. The code also implements source-head controls—a single shared mutation-effect head, a template-conditioned head, and a template-specific affine calibration on top of a shared head—to test whether the two template-specific sources need separate output calibration.

Architecture sensitivity analyses. Architecture controls include residual, latent, and mixture-style source-fusion variants, in which the source path produces a correction to the Tm prediction, a latent interaction between Tm and source features, or a gate between a target-only expert and a source-informed expert. The main reported FEP result uses the original shared-trunk architecture with separate template-specific source heads; the other variants are kept as sensitivity analyses, not as the main claim.

### Training objective, optimizer, and checkpoint selection

Each training example is a sequence, a scaled label, and a task identifier, and contributes loss only through the regression head for its task. The per-task loss is Huber loss with $\delta = 1.0$ on the scaled label. The default multi-task objective uses learned task-uncertainty weights (48): for task $k$ with loss $L_k$ and learned log-scale $s_k$, the contribution is

$$\frac{L_k}{2\exp(2s_k)} + s_k,$$

and the total loss sums the contributions from the tasks present in the minibatch. Fixed source-loss weights are also implemented and were included in the candidate searches for source classes where loss balance could shift validation performance.

Optimizer and schedule. The optimizer is AdamW. The default head/trunk learning rate is $3 \times 10^{-4}$, the default hot-encoder learning rate is $1 \times 10^{-4}$, weight decay is 0.04, dropout is 0.15, batch size is 16, warmup is 100 steps, and the schedule is cosine decay over at most 400 epochs. Checkpoints are evaluated and saved once per epoch, and mixed precision is used when CUDA is available. In the hot-encoder setting, the trainer creates separate optimizer parameter groups for ESM2 parameters and for the freshly initialized trunk/head parameters; in the frozen-encoder setting, ESM2 parameters are non-trainable and no encoder group is used.

Checkpoint selection. Within each run, the best checkpoint has the lowest mean squared error on the experimental Tm validation rows, and early stopping monitors the same metric with patience 15 and threshold 0. Source-task validation rows are not passed to the trainer in the production protocol. Thus, source labels shape the learned representation through the training loss, but they do not decide which checkpoint is used for target-task prediction.

### Candidate-setting selection and final evaluation

Candidate settings are selected on the experimental Tm validation split. The common grid includes the default setting; encoder learning rates of $3 \times 10^{-5}$, $1 \times 10^{-4}$, and $3 \times 10^{-4}$; a lower head learning rate of $1 \times 10^{-4}$ paired with encoder learning rate $3 \times 10^{-5}$; and dropout rates of 0.05, 0.15, and 0.30. Fixed source-loss weights are added for source classes where the source-label scale may need explicit balancing. Tm-only candidates include shared, residual, and latent architecture variants; mutation-effect candidates use the shared-trunk family for the main comparison, with a separate source-head design comparison; and MD-derived candidates include fixed-weight settings because Q-value labels differ in scale and interpretation from mutation free energies.

Seeded candidate evaluation. Each candidate is first evaluated as a three-seed ensemble on the Tm validation split, and the selected setting for each source class is then evaluated on the held-out test split as a ten-seed ensemble. Final comparisons use seeds 1–10. The run seed controls Python, NumPy, PyTorch, and CUDA randomness when CUDA is available, as well as target-training subsampling in the target-label scaling experiment, source-row sampling, and source train/validation splitting. The label-count curves use fixed source-specific training recipes while the number of source labels is varied.

### Statistics and stored outputs

The primary metric is MAE in degrees Celsius on the 396 held-out test examples. For each condition, the ten seeded models produce scaled Tm predictions for every test sequence; these are averaged in scaled-label space and inverse-transformed to degrees Celsius. Root mean squared error, coefficient of determination, and Spearman and Pearson correlations are also computed, but the figures emphasize MAE because it reads directly as an average temperature error.

Bootstrap intervals. Single-condition confidence intervals come from nonparametric bootstrap resampling over test examples. The absolute-error vector from the ten-seed ensemble is resampled with replacement 10,000 times using a fixed bootstrap seed, and the reported 90% interval is the 5th–95th percentile range of the bootstrapped MAE values. Paired comparisons use the same bootstrap indices for the candidate and reference absolute-error vectors. The reported $\Delta$MAE is candidate minus reference, so negative values mean lower error than the reference.

Stored run summaries. Each training and evaluation call writes a structured summary recording the command-line arguments, relevant environment variables, resolved model settings, per-label-count metrics, bootstrap intervals, per-example absolute errors, and paired bootstrap summaries. Compact tables derived from these summaries feed the main source comparison, the frozen-encoder controls, the source-head controls, and the source-combination controls.

### Interpreting the label-count curves

Target-label scaling reference. The target-label scaling curve is a positive control on the pipeline. It tests whether the same model family and evaluation code recover the expected gain when more true experimental Tm labels are available. The FEP source-label curve tests something different—whether more labels from a physically related but distinct computed quantity improve the same held-out Tm endpoint—while the MD Q-value curve tests whether simply adding more simulation-derived labels suffices when the source label measures a more indirect structural response. The relevant comparison is therefore not "simulation versus no simulation," but whether the source label carries physical information that transfers to the target property under target-centered model selection.

### Supplementary figure source tables

Supplementary figures are generated from tabular source files derived from the same run summaries as the main figures. The processed source tables, rendered supplementary figures, and figure-generation code will be distributed with the public analysis archive.

### Fig.-to-data and code mapping

The manuscript figures are regenerated from compact result summaries and source tables rather than from full training run directories. Table 2 summarizes the inputs used to audit or extend each figure.

Table 2. Manuscript Fig.-to-data mapping. The table identifies the processed summaries and scripts used to regenerate each display item.

| Display item | Inputs | Script | Primary comparison |
|---|---|---|---|
| Fig. 1 | Conceptual protocol | Main-figure script | Target-centered transfer-learning design |
| Fig. 2 | Main scaling summaries | Main-figure script | Label-count scaling for Tm labels, FEP labels, and MD Q-value labels |
| Fig. 3 | Final hot- and frozen-encoder source-screen summaries | Main-figure script | All-atom FEP and MD labels versus training only on experimental Tm labels |
| Fig. 4 | Final source-screen summary | Main-figure script | Rosetta, ThermoMPNN, and Rosetta-scored proposal sets |
| Supplementary Figs. 1–5 | Supplementary manifest and generated TSV tables | Supplementary-figure script | Data-set composition, candidate-setting searches, model controls, scaling controls, and MD-feature controls |

Generated supplementary tables. The final source-screen settings used in Figs. 3 and 4 and the frozen-encoder controls are recorded in generated supplementary tables, which list the selected architecture, validation-selected setting, source-label count, validation MAE, held-out test MAE, bootstrap interval, and run-summary reference for each condition. A panel-level mapping table for all supplementary figures is included in the analysis archive.

Supplementary Figs.

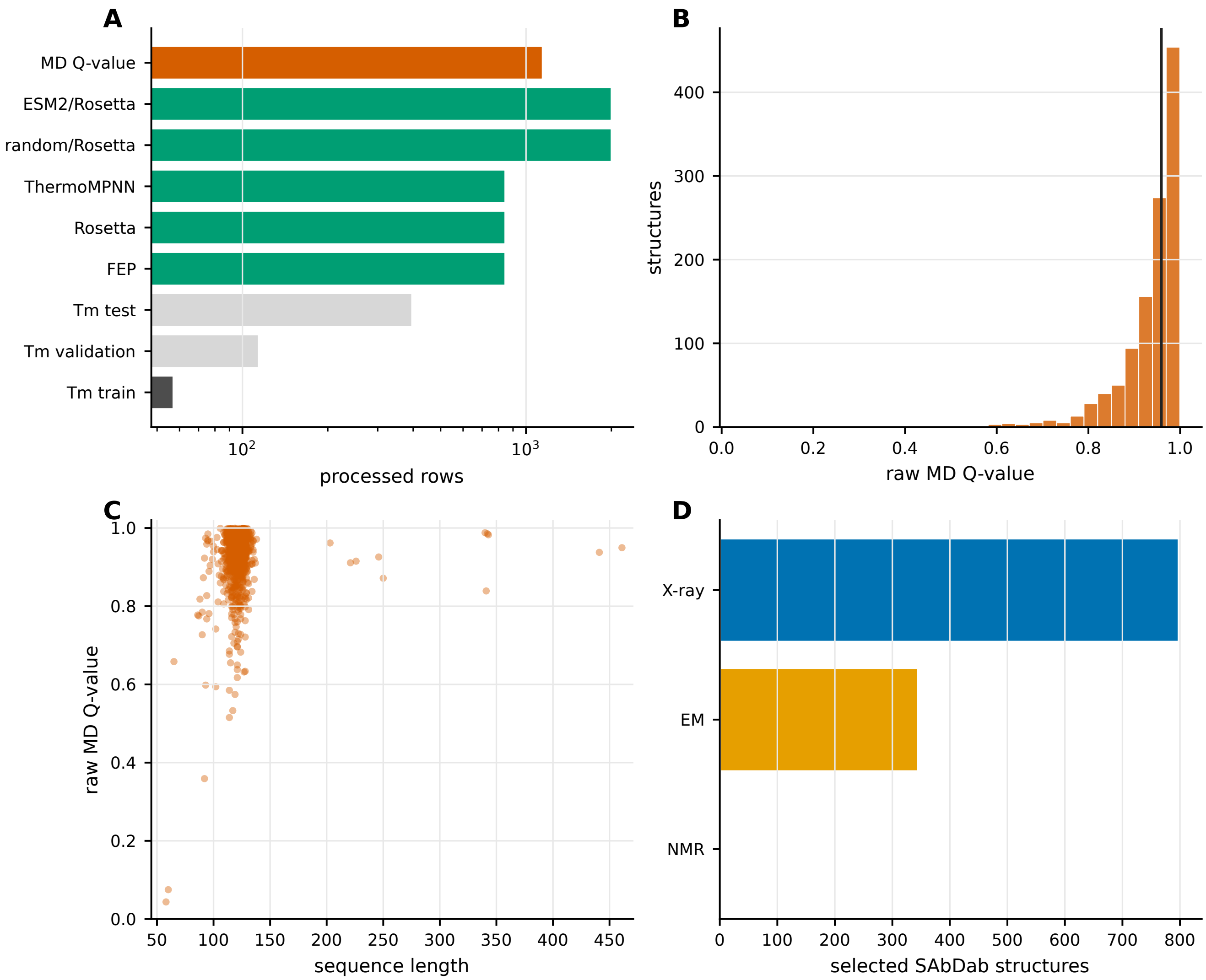


Fig. 5. Supplementary Fig. 1. Processed data sets and MD-derived Q-value labels. (a) Processed target and source-label row counts used in the reported experiments. (b) Distribution of raw MD-derived Q-values before min–max scaling. (c) Relationship between sequence length and raw MD-derived Q-value. (d) Experimental methods for the selected SAbDab structures used in the 400 K MD panel.

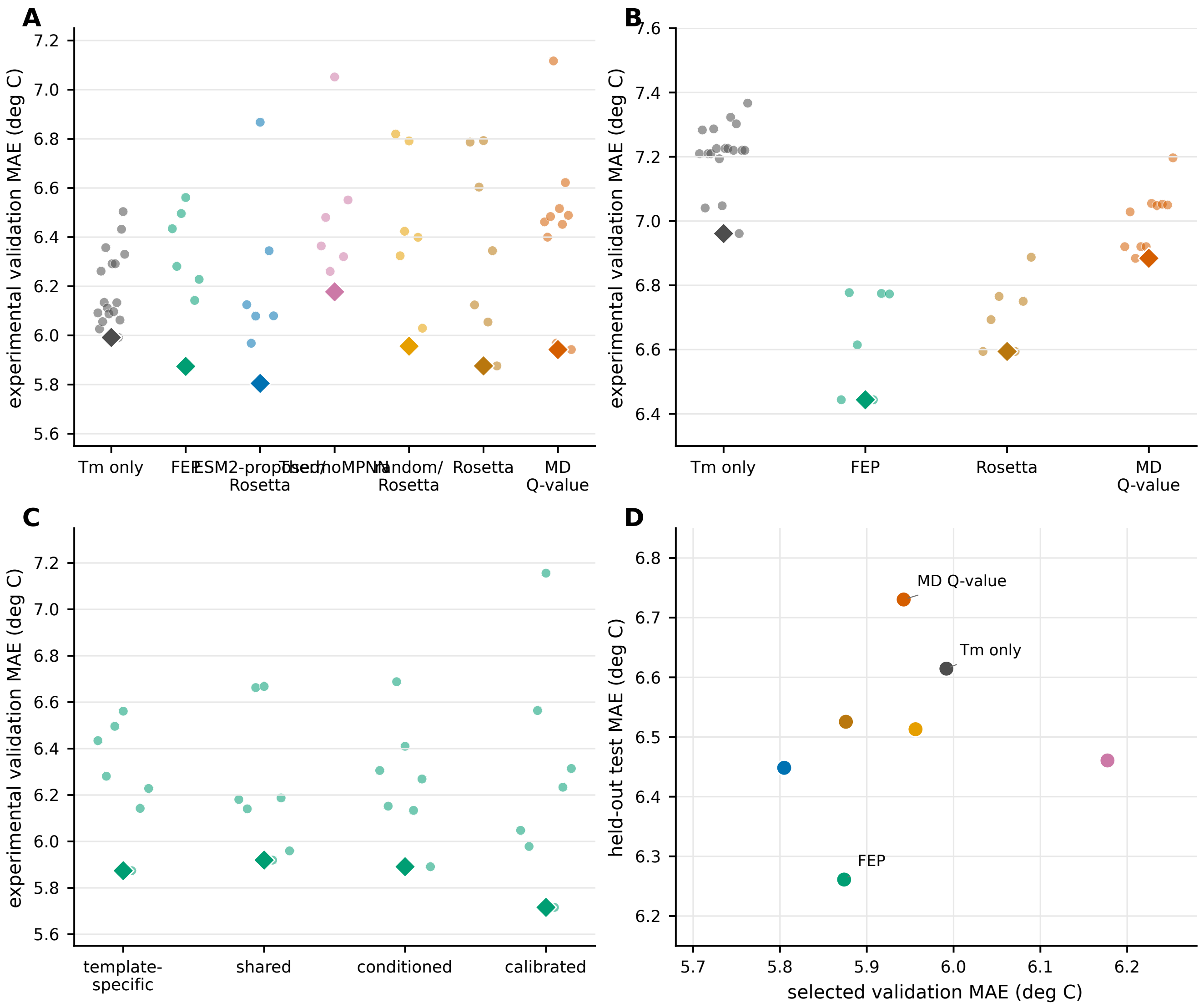


Fig. 6. Supplementary Fig. 2. Candidate-setting searches and target-validation selection. (a) Experimental validation MAE values for hot-encoder source comparisons. (b) Experimental validation MAE values for frozen-encoder controls. (c) Candidate source-head designs for FEP mutation free-energy labels. (d) Relationship between selected validation MAE and held-out test MAE for the final source comparison. Diamonds mark the best validation setting in panels a–c.

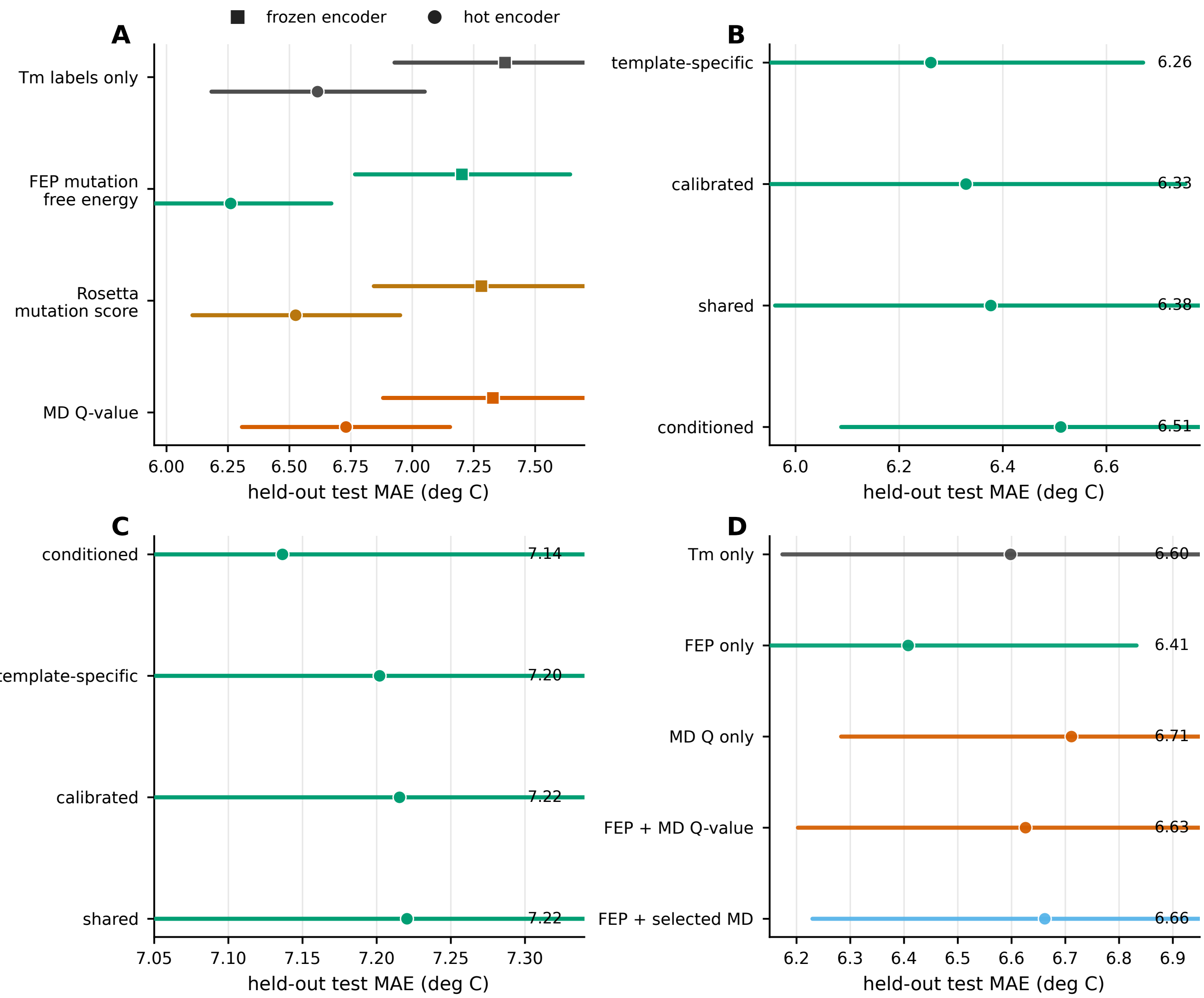


Fig. 7. Supplementary Fig. 3. Model and source-combination controls. (a) Frozen-encoder and hot-encoder comparisons for selected source labels. (b) Hot-encoder FEP source-head controls. (c) Frozen-encoder FEP source-head controls. (d) Source-combination controls comparing Tm-only, FEP-only, MD Q-value-only, and combined-source settings.

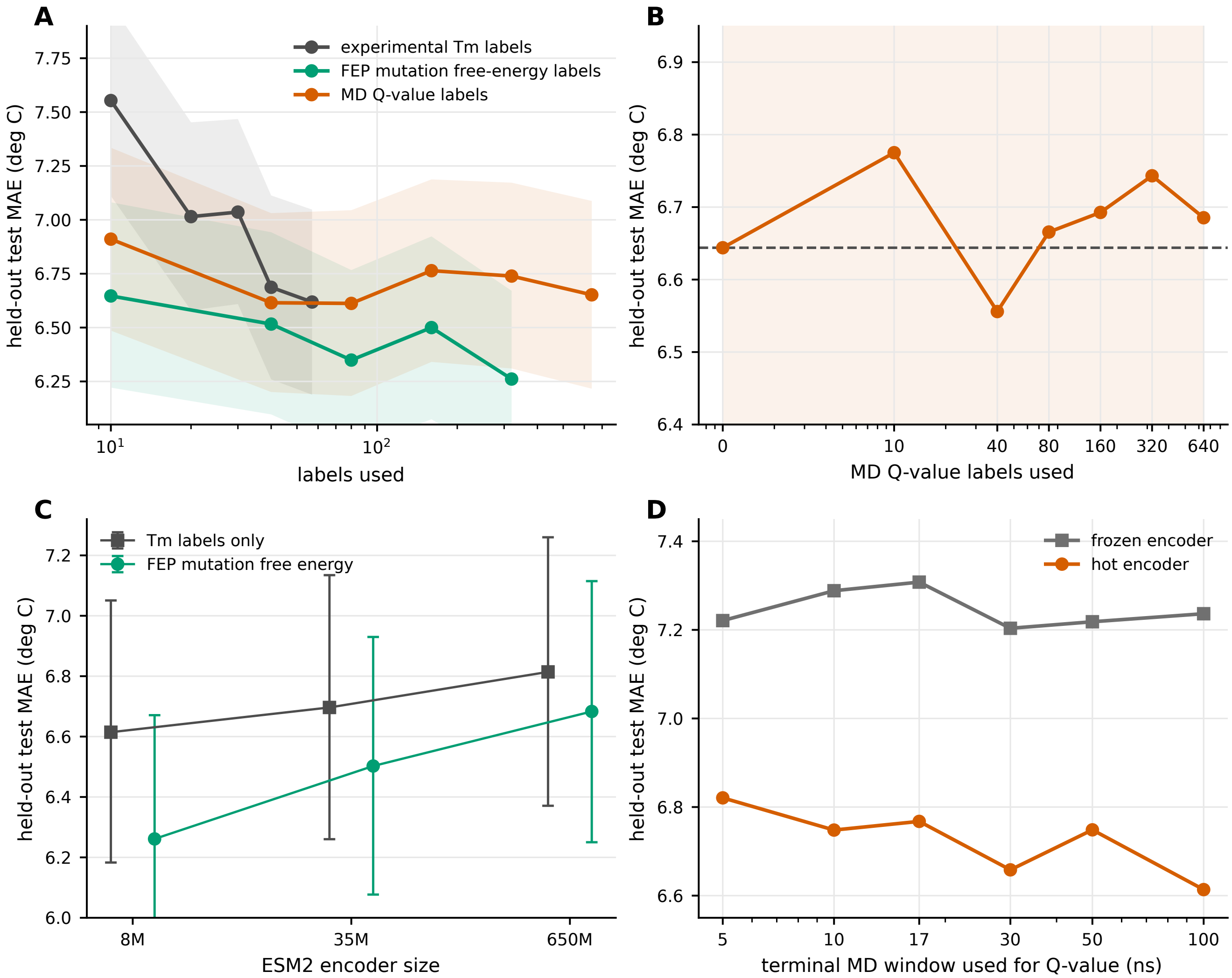


Fig. 8. Supplementary Fig. 4. Scaling, encoder-size, and MD-window controls. (a) Label-count curves for experimental Tm labels, FEP labels, and MD Q-value labels. (b) MD Q-value label-count curve after selecting the candidate setting separately for each label count on the experimental validation split. (c) ESM2 encoder-size controls for Tm-only and FEP settings. (d) MD trajectory-window controls for Q-value labels.

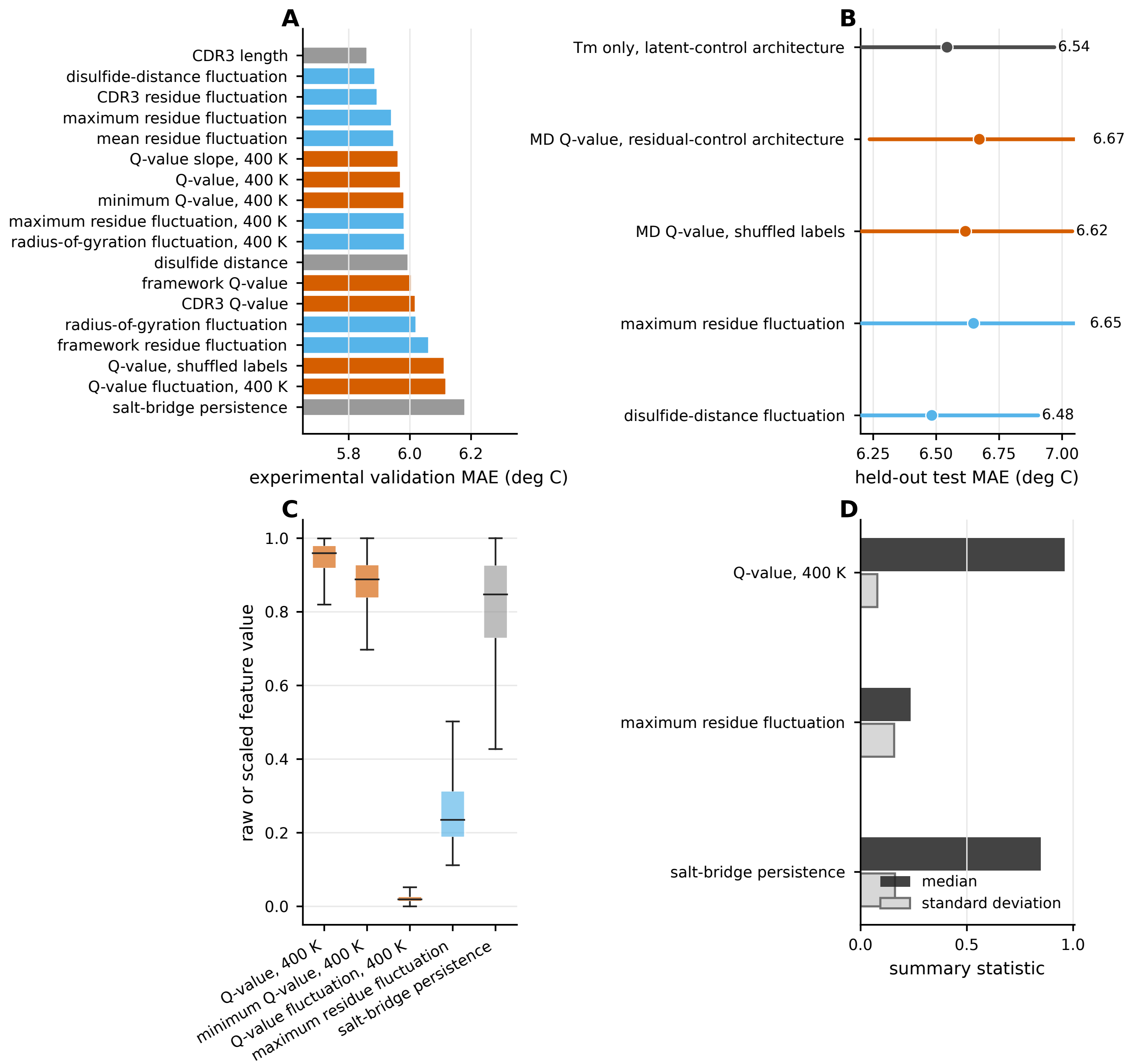


Fig. 9. Supplementary Fig. 5. MD feature and architecture controls. (a) Experimental validation MAE values for alternative MD-derived features. (b) Held-out test MAE values for selected architecture and shuffled label controls. (c) Distributions of representative MD-derived feature values. (d) Summary statistics for representative MD-derived feature values.